CASE STUDY / CONSULTING • DESIGN • SOFTWARE • IMPLEMENTATION

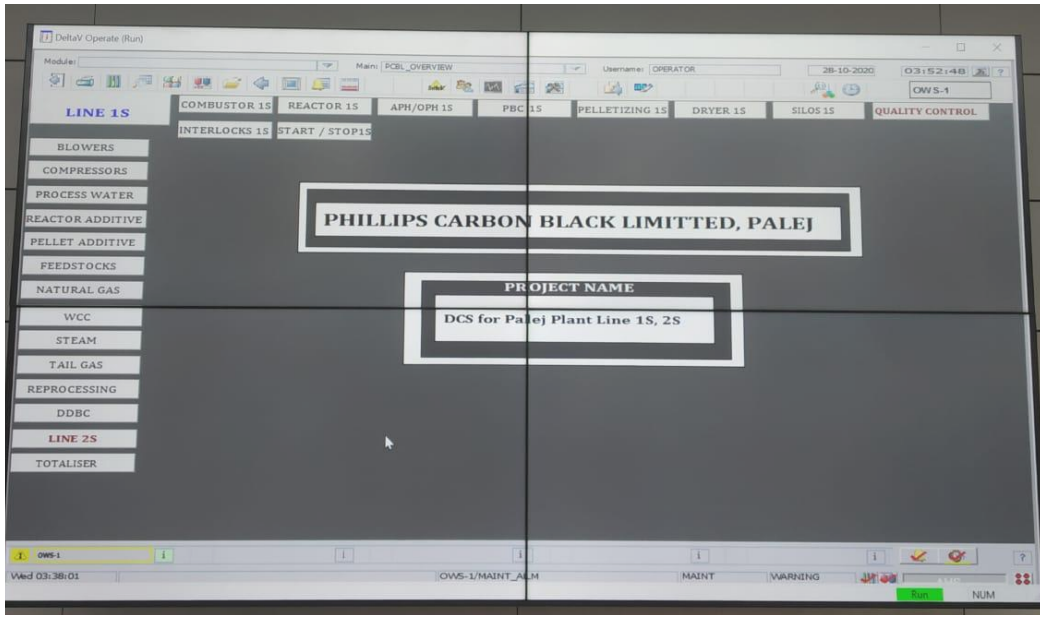
One plant view. Four connected displays.

DCS networking & video wall integration

Phillips Carbon Black Limited
Palej Plant, Gujarat

Samsung LCD displays, Kramer AV connectivity and D-TECH integration bring the plant's DCS graphics into a shared control-room view.

2 × 2	KRAMER	D-TECH
SAMSUNG LCD WALL	AV CONNECTIVITY	NETWORK & SOFTWARE



PROJECT VIEW | DELTAV GRAPHICS ON A 2 × 2 WALL



PCBL | PALEJ PLANT

Project photographs supplied by the client | Architecture and software views explain the integration design.

The requirement | A shared view of plant operations

The plant requirement is to make DCS information visible beyond an individual operator monitor, while preserving the existing control environment. A four-panel display surface provides a common view for the control-room team; video routing, network organisation and software controls make that view easier to select, manage and support.

01 **DISPLAY & CONNECTIVITY**

Samsung LCD wall, panel alignment, source resolution, Kramer signal routing and a documented output map.

02 **PLANT NETWORKING**

Labelled links, approved OT connectivity, separate AV management and controlled maintenance access.

03 **D-TECH SOFTWARE**

Source selection, permitted presets, device status, user permissions and an activity record.

04 **IMPLEMENTATION & HANDOVER**

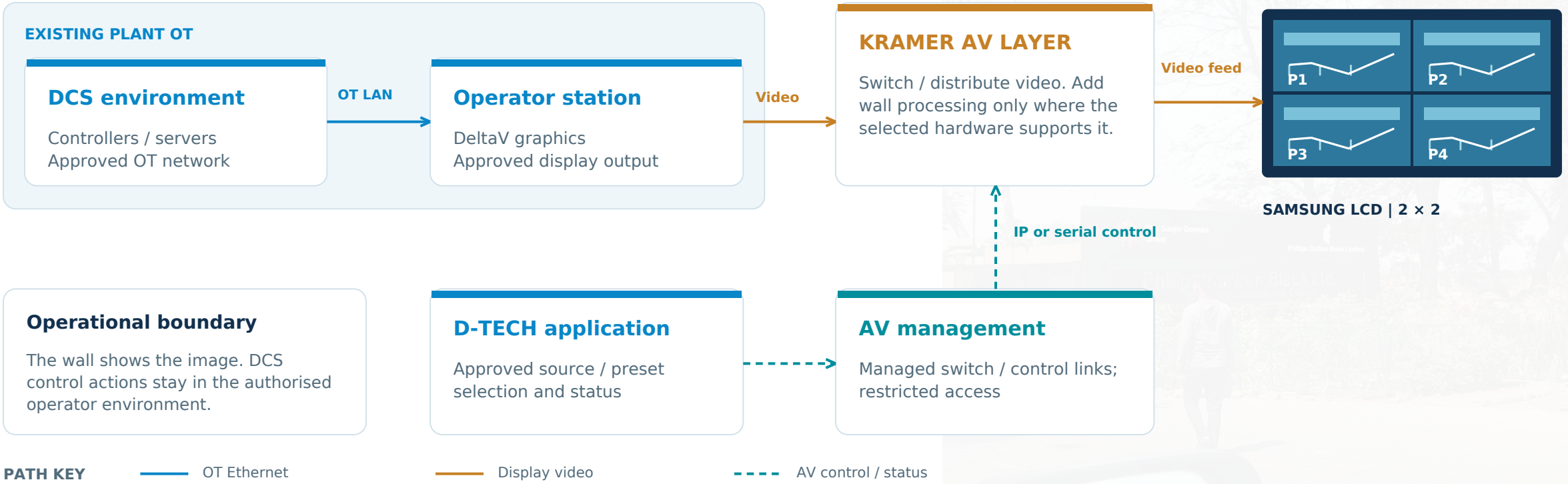
Survey, design approval, staging, mounting, configuration, site checks and support documentation.

Read the design

03 Architecture | 04 2D / 3D layout | 05 Network & software | 06 Deployment | 07 Verification | 08 Project record | 09 Client & support

Architecture | DCS picture, AV routing and software control

Existing DCS stations retain process-control authority. The video-wall layer receives an approved workstation display output.



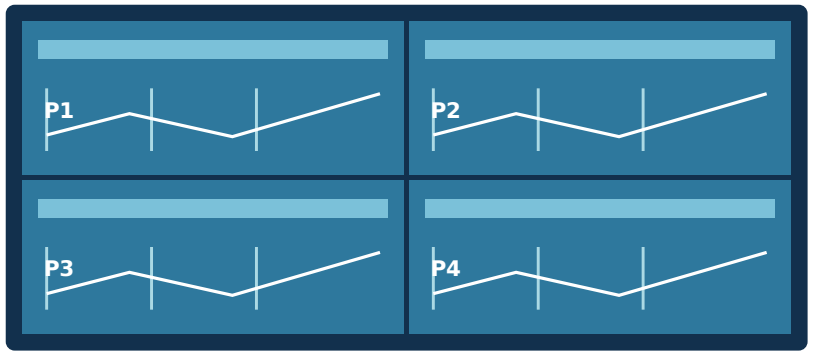
Logical design; source connectors, Kramer model, output topology and network details must match the approved equipment schedule.

2D FRONT ELEVATION / 3D CONCEPT ART / SERVICEABLE INSTALLATION

Display design | Four panels, one planned viewing surface

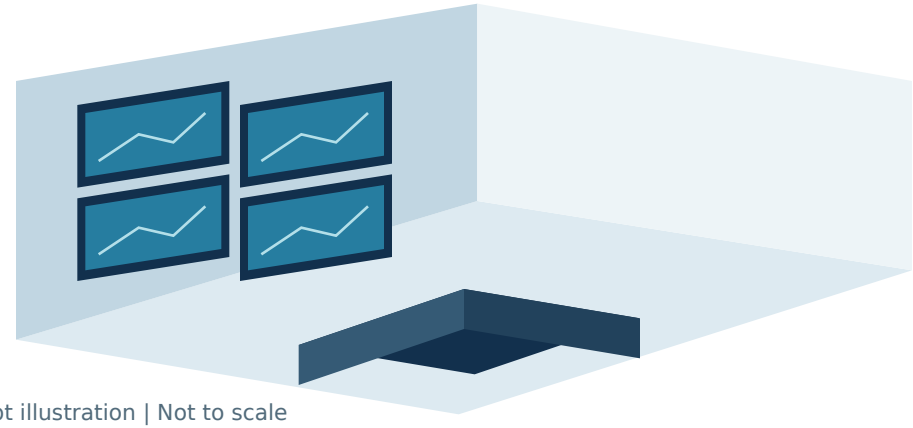
The supplied photographs show a 2 × 2 arrangement. The sketches explain panel positions and the physical integration concept.

2D FRONT ELEVATION



Overall width = 2 × panel width + joint

3D CONTROL-ROOM CONCEPT



Concept illustration | Not to scale

ALIGN & CALIBRATE

Level the mount grid; match colour and brightness; verify bezel compensation and screen order.

CABLE & POWER

Label each video output and power feed. Size supports, UPS and circuits against actual equipment loads.

MAINTAIN & ACCESS

Follow OEM ventilation and service clearances. Confirm front or rear access for the selected mounts.

A matrix routes sources; multi-window composition or arbitrary layouts require a suitable processor. The installed models determine the available functions.

Network & software | Controlled access, clear device status

The D-TECH software layer is designed around the AV equipment interface. Its role is to simplify approved display operations and make source, preset and device status understandable to the user.

D-TECH | WALL CONTROL

DESIGN WIREFRAME

SOURCE	DCS operator display
PRESET	Plant overview
STATUS	Confirmed / unknown
ACCESS	Approved user role

Select preset → validate → send command → confirm → log

Software module	Development responsibility
User interface	Source catalogue, preset selection and status view.
Device adapter	Supported Kramer IP / serial commands, replies and timeouts.
Roles & history	Viewer, AV operator and administrator; user / time / route log.
Optional dashboard	Client-approved, licensed read-only DCS data interface; separate scope.

Network zone	Design intent
PLANT OT	Preserve the approved DCS topology; connect only authorised operator / interface nodes.
AV MANAGEMENT	Separate management access to Kramer and supported display controls; allow only required endpoints.
OPTIONAL DATA INTERFACE	Use an approved gateway / DMZ route for dashboards. No application process-control writeback.

Software acceptance checks: reject unauthorised changes, confirm device replies, show unknown status on timeout and retain configuration backups.

Implementation | Survey, develop, deploy and hand over

D-TECH coordinates the physical display work, networking and software with the plant IT / instrumentation team and the approved DCS interface owner.

01 SURVEY & DISCOVER

Map sources, connectors, viewing position, cable paths, loads and maintenance access.

02 DESIGN & APPROVE

Agree the mount plan, signal map, OT / AV boundaries, software functions and test criteria.

03 STAGE & DEVELOP

Verify panels and routing; build device adapters, user roles, presets and configuration backups.

04 MOUNT & CONNECT

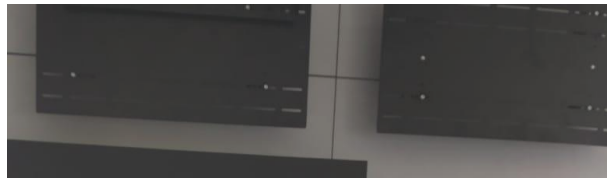
Install supports and displays; label video / network / power links and register device details.

05 CONFIGURE & TEST

Apply screen positions, matching settings and routes; check permissions, recovery and display quality.

06 HANDOVER & SUPPORT

Provide drawings, asset list, settings, software release notes, training and the support route.



MOUNTING GRID | SITE PHOTO



ASSEMBLED LCD WALL

Commissioning | Verify the complete display chain

Acceptance should be recorded against the approved design, with plant representatives witnessing the checks that affect DCS display availability.

Check	Acceptance record to capture
SOURCE & PANEL MAP	Correct DCS source appears on P1-P4; no missing quadrant, duplicated area or incorrect orientation.
READABILITY & GEOMETRY	Plant graphics, labels and alarm colours remain readable from agreed operator positions; scaling and bezel mapping recorded.
VIDEO STABILITY	Approved resolution and refresh rate; no intermittent signal loss during the agreed soak period.
SOFTWARE & PERMISSIONS	Permitted presets work; unauthorised changes fail; commands, acknowledgements and timeout states are logged.
NETWORK & RECOVERY	Approved access rules; cable / power interruption behaviour checked; recovery and fallback procedure recorded.
DOCUMENTS & HANDOVER	Asset / serial register, port map, settings backup, software version, user guide and signed acceptance sheet.

Support process

Log issue → identify source / route / panel → diagnose → restore display → verify with plant → close with notes

These are commissioning criteria and the support method; the supplied material contains no signed test results or measured uptime / response-time commitments.

Project record | Photographs, references and design basis

The case study brings the site photographs and client-described solution together with a clear architecture and delivery method. The reference register identifies what each source establishes.

Reference	What it establishes
CLIENT BRIEF	Networking and DCS video-wall integration; Samsung LCD displays, Kramer connectivity and D-TECH software scope.
PROJECT PHOTOGRAPHS	Wall-mount installation stages, a four-panel 2 × 2 array and DeltaV / PCBL Palej graphics displayed on the wall.
PO 4700014044 | 20 Jan 2020	Related PCBL Palej CCTV / networking supply reference: NVR, PoE switch, IP cameras, cable and installation accessories.
PO 4500095327 | 23 Jan 2020	Related PCBL Palej storage-supply reference.
PRIMARY OEM REFERENCES	Samsung display configuration guidance; Kramer device-control protocol; Emerson example of approved read-only data access.

Design confirmation before replication or expansion

The supplied POs do not specify the Samsung / Kramer wall bill of materials. Confirm LCD and Kramer models, connector / output map, software release and DCS-interface approval against the final equipment schedule and as-built records. The logical drawings and software functions describe the design; they do not replace those records.

Client profile | Plant integration with lifecycle support

Phillips Carbon Black Limited

PCBL Palej Plant
National Highway No. 8, Palej - 392220
District Bharuch, Gujarat, India



CLIENT PLANT | PALEJ ENTRANCE

D-TECH integration responsibility

Consulting and architecture, display / network coordination, software development, implementation planning, commissioning documentation and ongoing service support.

OPEN D-TECH SUPPORT PORTAL

+91 99989 03042 | AMC@DTECHINDIA.COM

Design references

Samsung video-wall support / manuals | Kramer Protocol 3000
Emerson read-only integration reference